DESIGN AND IMPLEMENTATION OF A SMART DOORBELL WITH GSM-BASED REMOTE COMMUNICATION USING ESP32

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DEGREE IN PHYSICS

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I, ALAMUTU YASIR OLAMILEKAN hereby declare that the project work titled DESIGN AND IMPLEMENTATION OF A SMART DOORBELL USING GSM-BASED REMOTE COMMUNICATION USING ESP32 is a record of an original work done by me, as a result of my research effort carried out in the Faculty of Physical and Earth Sciences, University of Lagos, Nigeria.

Under the Supervision of _____

_____

Students Signature & Date

CERTIFICATION

This is to certify that this study was carried out by ALAMUTU YASIR OLAMILEKAN, matric Number 210808002 in the Department of PHYSICS, Faculty of PHYSICAL AND EARTH SCIENCES, UNIVERSITY OF LAGOS, NIGERIA, in partial fulfillment of the requirements for the award of Bachelor of Science (B.Sc. Hons) Degree in Physics

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DEDICATION

This project is dedicated to Almighty ALLAH, whose grace, wisdom, strength, and guidance have made the successful completion of this work possible.

I also dedicate this work to my parents and family for their unwavering love, support, encouragement, and sacrifices throughout my academic journey. Their belief in me has been a constant source of motivation and inspiration.

I sincerely appreciate and acknowledge my uncle for his generous financial support throughout the course of my academic journey. His willingness to assist whenever the need arose greatly eased many of the challenges associated with my education and this project. His encouragement, sacrifices, and belief in my potential served as a source of motivation and inspiration. I am deeply grateful for his kindness and unwavering support, which contributed significantly to the successful completion of this work.

Finally, this work is dedicated to all students and researchers in the field of Physics, Electronics, and Embedded Systems, whose pursuit of knowledge and innovation continues to contribute to technological advancement and societal development.

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CHAPTER ONE

INTRODUCTION

1.1 Background to the Study

Communication is a fundamental aspect of human existence and plays a crucial role in facilitating interaction within various environments such as homes, offices, industries, and security installations. The evolution of communication systems has significantly transformed the manner in which information is transmitted, progressing from traditional face-to-face interactions to sophisticated electronic and wireless communication networks (Rappaport, 2002; Stallings, 2013). Among these systems, intercom devices have been widely adopted as a means of enabling direct voice communication between individuals located in different areas within a defined environment. Traditional intercom systems are primarily based on wired communication. These systems involve the use of physical cables to connect communication endpoints, allowing users to exchange voice signals over short distances. Wired intercom systems are commonly used in residential buildings, office complexes, hospitals, and access control systems where communication between entry points and control units is necessary. Despite their usefulness, these systems are associated with several inherent limitations.

One major limitation of wired intercom systems is their restricted range of operation. Communication is confined to the physical length of the connecting cables, making them unsuitable for long-distance communication. Additionally, the installation process of wired systems can be complex and costly, particularly in large buildings or areas where infrastructure is not already in place. The need to lay cables through walls, ceilings, or underground pathways increases both labor and material costs.

Furthermore, wired intercom systems lack flexibility and scalability. Once installed, modifying or expanding the system requires additional wiring, which may not always be feasible. Maintenance of such systems can also be challenging, especially when faults occur within concealed wiring. These limitations highlight the need for more efficient and adaptable communication solutions. The emergence of wireless communication technologies has provided a viable alternative to traditional wired systems. Wireless communication eliminates the need for physical connections, allowing devices to communicate over long distances through electromagnetic waves. Among the various wireless communication technologies available, the Global System for Mobile Communication (GSM) stands out as one of the most widely used and reliable systems. GSM is a digital cellular network standard that enables voice and data communication over long distances. It operates through a network of base stations and switching systems, allowing users to communicate across different geographical locations. GSM technology has been widely adopted due to its extensive coverage, reliability, and relatively low cost of implementation. It provides an effective platform for developing communication systems that are not constrained by physical distance.

In addition to advancements in communication technology, the development of embedded systems has played a significant role in modern electronic design. Embedded systems are specialized computing systems developed to perform dedicated functions within larger electronic systems, often operating under real-time constraints (Mazidi et al., 2014). These systems typically consist of microcontrollers, sensors, and other electronic components that work together to achieve a desired function. The ESP32 microcontroller is a highly integrated embedded platform that combines processing capability, wireless communication functionality, and low-power operation

in a single device (Espressif Systems, 2020). It offers features such as dual-core processing, low power consumption, and multiple communication interfaces including UART, I2C, and SPI. These features make the ESP32 suitable for integration with communication modules, sensors, displays, and other peripheral devices in embedded applications (Espressif Systems, 2020) When combined with GSM modules such as the SIM800L, embedded systems can be used to develop intelligent communication devices capable of performing tasks such as making calls, receiving calls, and sending messages. The SIM800L module is a compact quad-band GSM/GPRS communication module capable of voice communication, SMS transmission, and data communication through AT command interfaces (SIMCom Wireless Solutions, 2013).

This project focuses on the design and implementation of a GSM-based intercom system using the ESP32 microcontroller and SIM800L module. The system is designed to replace conventional wired intercom systems with a wireless alternative that provides greater flexibility, portability, and ease of deployment. By leveraging GSM technology, the system enables long-distance voice communication without the need for physical wiring. The integration of a microphone and speaker allows for real-time audio communication, while an LCD provides visual feedback on system status. A portable power supply system ensures that the device can operate independently of a fixed power source, making it suitable for various applications.

1.2 Statement of the Problem

Despite the widespread use of intercom systems, traditional designs are associated with several limitations that reduce their effectiveness in modern applications. One of the primary challenges is the dependence on wired infrastructure. Wired intercom systems require extensive cabling to connect communication points, which increases installation complexity and cost. In large buildings

or remote locations, laying cables can be impractical and time-consuming. The need for physical connections also limits the flexibility of the system, making it difficult to relocate or expand communication points without significant modifications. This lack of scalability is a major drawback in environments where communication needs may change over time.

Another challenge is maintenance. Faults in wired systems can be difficult to detect and repair, especially when cables are hidden within walls or underground. This can lead to increased downtime and additional maintenance costs. Furthermore, conventional intercom systems are limited in range and cannot support long-distance communication beyond the physical boundaries of the installed infrastructure. In situations where communication is required across different locations, these systems become ineffective.

Although mobile phones provide long-distance communication, they are not specifically designed for dedicated intercom applications. They lack integration with embedded control systems and cannot be easily incorporated into automated communication setups. There is therefore a need for a communication system that overcomes these limitations by eliminating the dependence on physical wiring, providing long-distance communication capability, and offering flexibility and ease of deployment. Such a system should also be cost-effective, portable, and capable of supporting real-time voice communication.

1.3 Aim and Objectives of the Study

The aim of this project is to design and implement a GSM-based intercom system using an ESP32 microcontroller. The objectives of this study are to:

(i) Design a wireless intercom system that utilizes GSM technology for communication.

(ii) Interface the SIM800L GSM module with the ESP32 microcontroller.

(iii) Implement two-way voice communication using a microphone and speaker.

(iv) Develop a user interface using an LCD to display system status.

(v) Design a stable and portable power supply system suitable for GSM operation.

(vi) Integrate all components into a compact and functional hardware system.

(vii) Test and evaluate the performance of the system under real operating conditions.

1.4 Significance of the Study

This study is significant in several ways. First, it provides a practical solution to the limitations of traditional wired intercom systems. By utilizing GSM technology, the system eliminates the need for physical connections and allows communication over long distances. Second, the project demonstrates the integration of embedded systems with wireless communication technologies. This contributes to the development of modern communication devices that are flexible, efficient, and cost-effective. Third, the system offers a portable communication solution that can be deployed in various environments, including residential buildings, offices, security systems, and remote locations. The use of a rechargeable power supply enhances its applicability in areas where stable power is not readily available. Additionally, the project provides a platform for further research and development in the field of embedded systems and communication engineering. It can be expanded to include additional features such as remote monitoring, automation, and internet connectivity.

1.5 Scope of the Study

This project is focused on the design and implementation of a GSM-based intercom system capable of making and receiving voice calls. The system includes the following components:

- ESP32 microcontroller for system control
- SIM800L GSM module for communication
- Microphone and speaker for audio input and output
- LCD for displaying system status
- Rechargeable battery and voltage regulation system

The system operates within the coverage area of a GSM network and is designed for basic intercom functionality. Advanced features such as video communication, internet-based communication, and large-scale networking are not included in this study.

1.6 Definition of Terms

GSM (Global System for Mobile Communication): A digital cellular network used for voice and data communication.

ESP32: A microcontroller used in embedded systems with built-in communication interfaces.

SIM800L: A GSM module used for making calls and sending SMS messages.

Intercom System: A communication device that allows voice interaction between different locations.

Embedded System: A specialized computing system designed to perform specific functions.

Microcontroller: A compact integrated circuit used to control electronic systems.

Wireless Communication: Transmission of information without the use of physical connections.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter presents a review of relevant literature and existing technologies related to communication systems, particularly wireless intercom systems. It also discusses GSM communication, embedded systems, audio interfacing, and power management techniques, which form the foundation of this project. The purpose of this chapter is to provide a theoretical background and highlight previous works that relate to the development of a GSM-based intercom system.

2.2 Intercom Systems

Intercom systems are electronic communication devices that enable voice communication between two or more locations. These systems are widely used in residential buildings, offices, hospitals, and security installations. Traditional intercom systems are based on wired communication, where communication points are connected through physical cables.

According to O'Reilly (2018), wired intercom systems are reliable and provide clear audio communication over short distances. However, they are limited by installation complexity, high cost, and lack of flexibility. In large buildings or remote locations, the installation of cables becomes difficult and expensive. Wireless intercom systems have been developed to overcome these limitations. These systems use technologies such as radio frequency (RF), Wi-Fi, and GSM to facilitate communication without physical connections. Among these technologies, GSM is

particularly advantageous due to its wide coverage and ability to support long-distance communication.

2.3 GSM Communication Technology

The Global System for Mobile Communication (GSM) is a widely used digital cellular network that enables voice and data transmission. GSM operates using a combination of Time Division Multiple Access (TDMA) and Frequency Division Multiple Access (FDMA), which allows multiple users to share the same communication channel efficiently (Rappaport, 2002). GSM networks consist of base stations, switching centers, and mobile devices that work together to provide seamless communication across large geographical areas. One of the major advantages of GSM technology is its extensive coverage, which allows communication even in remote locations. GSM modules such as the SIM800L are compact devices that enable embedded systems to access GSM networks. These modules support voice calls, SMS, and data communication through AT command instructions (SIMCom, 2013). The use of GSM modules in intercom systems eliminates distance limitations and allows communication over long distances without additional infrastructure.

2.4 Embedded Systems and Microcontrollers

Embedded systems are specialized computing systems designed to perform specific functions within a larger system. They are widely used in modern electronic devices, including communication systems, automation systems, and control systems. Microcontrollers are the core components of embedded systems. They provide processing capability, input/output control, and communication interfaces. The ESP32 microcontroller is one of the most advanced

microcontrollers available today, offering dual-core processing, integrated Wi-Fi and Bluetooth connectivity, low power consumption, and support for multiple communication protocols such as UART, SPI, and I²C. The ESP32 is particularly suitable for this project due to its ability to interface with GSM modules and peripheral devices such as LCDs. It serves as the control unit of the system, managing communication and user interaction **(Espressif Systems, 2020).**

2.5 GSM Integration in Embedded Systems

The integration of GSM modules into embedded systems enables remote communication capabilities. GSM modules communicate with microcontrollers using serial communication protocols, typically through UART. According to Mazidi et al. (2014), GSM modules operate using AT commands, which allow microcontrollers to control functions such as dialing, answering calls, and sending SMS. This command-based interface simplifies the integration process and makes GSM modules suitable for various applications. In this project, the SIM800L module is used to provide GSM functionality. The ESP32 sends AT commands to the module to initiate calls and manage communication. This integration enables the system to function as a wireless intercom device.

2.6 Audio Communication in GSM Systems

Audio communication in GSM systems involves the transmission of analog voice signals through digital networks. GSM modules provide dedicated pins for microphone input and speaker output, allowing direct interfacing with audio components. The SIM800L module includes built-in audio processing features such as noise reduction and echo cancellation, which improve communication quality (SIMCom, 2013). The microphone captures voice signals, which are converted into

electrical signals and transmitted through the GSM network. On the receiving end, the module outputs the audio signal to a speaker. This process enables real-time two-way communication, which is essential for intercom applications.

2.7 Liquid Crystal Display (LCD) in Embedded Systems

Liquid Crystal Displays (LCDs) are widely used in embedded systems to provide visual feedback. They are preferred due to their low power consumption, ease of interfacing, and ability to display text information. According to Monk (2016), LCD modules can be interfaced using I2C communication, which reduces the number of required pins and simplifies circuit design. In this project, the LCD is used to display system status such as “System Ready,” “Dialing,” and “Call Active.”

2.8 Power Supply Considerations in GSM Systems

Power supply is a critical aspect of GSM-based systems. GSM modules require a stable power source capable of handling high current bursts during transmission. According to SIMCom (2013), the SIM800L module operates within a voltage range of 3.4V to 4.4 V and can draw peak currents of up to 2A. Improper power supply design can lead to system instability, communication failure, and module resets. To ensure stable operation, voltage regulation components such as buck converters and capacitors are used. Rechargeable battery systems are also employed to provide portability and reliability.

2.9 Review of Related Works

Several researchers have investigated the application of GSM technology in communication, security, monitoring, and automation systems.

Akinwale et al. (2019) developed a GSM-based security alert system capable of sending notifications to users whenever unauthorized access was detected. Their system demonstrated the effectiveness of GSM technology for remote communication and security monitoring.

Kumar and Patel (2017) designed a GSM-based industrial monitoring system that enabled remote supervision of industrial equipment through wireless communication. Their study showed that GSM networks could provide reliable communication over long distances without requiring dedicated infrastructure.

Al-Fuqaha et al. (2015) discussed the integration of communication technologies within embedded systems and Internet of Things (IoT) environments. Their work highlighted the importance of wireless communication technologies in modern intelligent systems.

Yick, Mukherjee, and Ghosal (2008) reviewed wireless communication systems used in sensor network applications and demonstrated the advantages of wireless communication in reducing installation complexity and improving scalability.

Xia et al. (2012) examined the development of intelligent communication systems based on wireless technologies and emphasized the growing importance of embedded communication devices in modern engineering applications.

Although previous studies have demonstrated the effectiveness of GSM technology in security systems, industrial monitoring systems, and embedded communication applications, relatively few studies have focused on the implementation of GSM-based intercom systems capable of supporting dedicated real-time two-way voice communication. Therefore, this study addresses this gap by developing a GSM-based intercom system using the ESP32 microcontroller and SIM800L GSM module to provide a flexible and portable wireless communication solution.

CHAPTER THREE

METHODOLOGY

3.1 Introduction

This chapter describes the design and implementation of the GSM-based intercom system. It includes the system architecture, hardware components, circuit design, and software development. The aim is to provide a detailed explanation of how the system was developed and how its components interact.

3.2 System Overview

The system is designed as a wireless intercom device that enables voice communication using GSM technology. It consists of an ESP32 microcontroller, SIM800L GSM module, microphone, speaker, LCD display, and a power supply system.

The ESP32 serves as the control unit, while the SIM800L provides communication functionality. The microphone and speaker handle audio input and output, and the LCD provides user feedback.

3.3 System Block Diagram

The system can be divided into the following functional blocks:

- Input unit (microphone and push button)
- Processing unit (ESP32 microcontroller)
- Communication unit (SIM800L GSM module)
- Output unit (speaker and LCD)
- Power supply unit

3.4 Hardware Design

3.4.1 ESP32 Microcontroller

The ESP32 is used as the main control unit of the system. It processes input signals, sends commands to the GSM module, and controls the LCD display. The ESP32 was selected due to its high processing capability, integrated communication interfaces, low power consumption, and ease of integration with external communication modules (Espressif Systems, 2020).

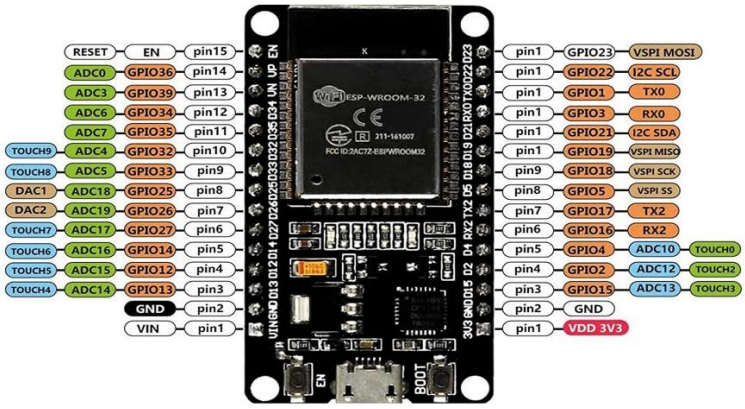


Figure 3.1: Diagram of ESP32 Microcontroller

3.4.2 SIM800L GSM Module

The SIM800L module is responsible for communication. It is used to make calls and transmit voice signals. It operates using AT commands sent from the ESP32. The SIM800L module was selected because of its compact size, low power requirements, and support for voice communication through GSM networks (SIMCom Wireless Solutions, 2013).



Figure 3.2: Diagram of SIM800L GSM Module

3.4.3 Audio Interface

The audio interface of the system consists of a telephone-style intercom handset containing an inbuilt microphone and speaker. The microphone was connected to the **MIC+** and **MIC−** pins of the SIM800L module to capture voice input, while the speaker was connected to the **SPK+** and **SPK−** pins for audio output during calls. The handset enabled real-time two-way voice communication without the need for an external amplifier, making the system simpler, more compact, and power efficient.



Figure 3.3: Diagram of Telephone Handset Used as Audio Interface

3.4.4 LCD Display

The LCD is used to display system status messages. It is interfaced with the ESP32 using the I2C communication protocol. The LCD display was interfaced using the I2C protocol to reduce wiring

complexity and improve communication efficiency between the display unit and the ESP32 microcontroller (Monk, 2016).



Figure 3.4: Diagram of LCD Display Unit

3.4.5 Power Supply Unit

The power supply consists of a rechargeable battery, power module, and buck converter. The SIM800L is powered with a regulated 4V supply, while the ESP32 and other components are powered with 5V.

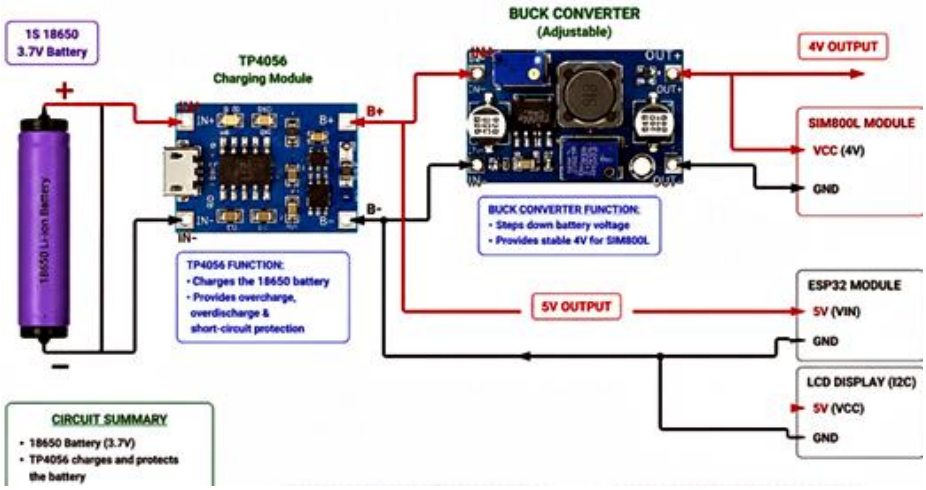


Figure 3.5: Diagram of Power Supply Unit

3.5 Circuit Design

The circuit is designed using a vero board. All components are connected according to their functional requirements. The SIM800L is connected to the ESP32 via UART communication, while the LCD is connected using I2C. Proper grounding and voltage regulation are ensured to maintain system stability.

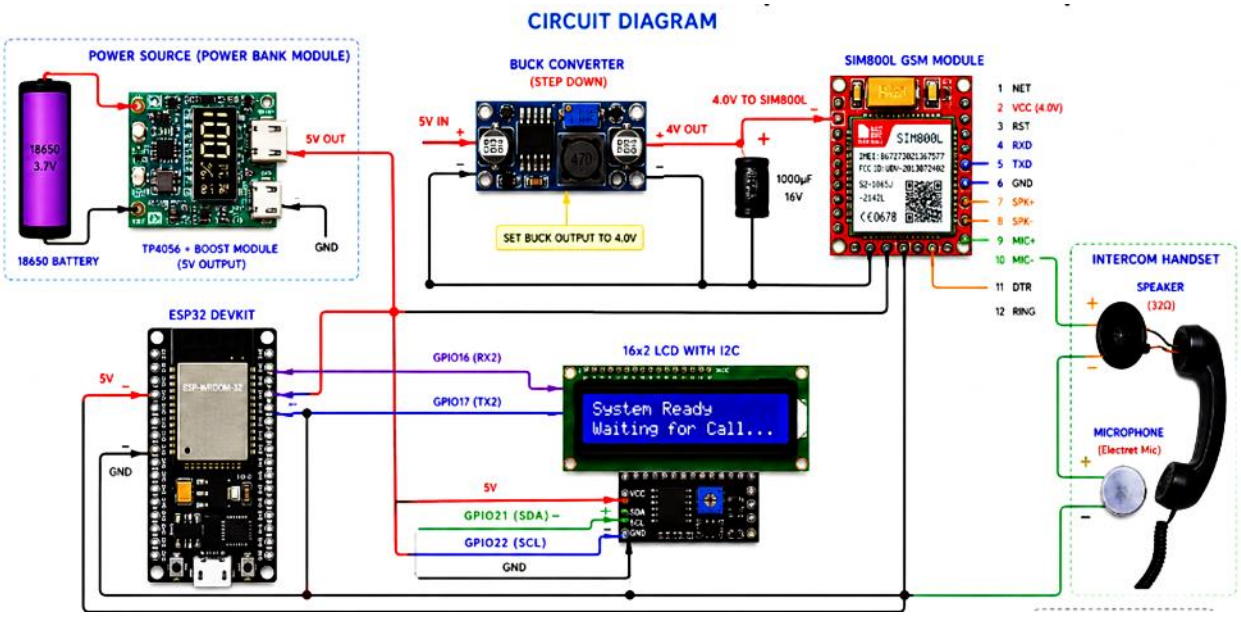


Figure 3.6: Circuit Diagram of the Complete GSM-Based Intercom System

3.6 Software Design

The software is developed using the Arduino IDE. The ESP32 is programmed to send AT commands to the SIM800L module to initiate calls when a button is pressed. The system logic ensures that the call is initiated once and remains active until the receiver ends the call.

3.7 System Operation

When the button is pressed, the ESP32 sends a command to the SIM800L to dial a predefined number. Once the call is connected, the user can communicate through the microphone and speaker. The call remains active until the receiver ends it, ensuring that only the owner has control over call termination.

CHAPTER FOUR

RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the results obtained from the design and implementation of the GSM-based intercom system. It discusses the performance of the developed hardware and software, evaluates the functionality of the major components, and analyzes the effectiveness of the system during testing. The chapter also explains the operational reliability of the developed system and its ability to achieve the project objectives.

4.2 System Implementation

The GSM-based intercom system was successfully designed and implemented using the ESP32 microcontroller and the SIM800L GSM module. The completed system was assembled on a vero board and packaged inside a square-shaped enclosure together with a telephone-style handset for communication.

The implemented system consists of:

- ESP32 microcontroller
- SIM800L GSM module
- LCD display unit
- Push button
- Rechargeable power supply unit
- TP4056 charging module
- Buck converter

- Telephone handset microphone and speaker

The system was designed to operate wirelessly using GSM technology. When the push button is pressed, the SIM800L module automatically places a call to the predefined phone number programmed into the ESP32.

4.3 Hardware Performance Evaluation

4.3.1 ESP32 Microcontroller Performance

The ESP32 functioned effectively as the main control unit of the system. It successfully coordinated all system activities, including:

- Button detection
- LCD control
- GSM communication
- AT command processing

The microcontroller maintained stable serial communication with the SIM800L module throughout the testing process.

The ESP32 also provided fast response time and reliable operation during continuous testing.

4.3.2 SIM800L GSM Module Performance

The SIM800L GSM module successfully connected to the GSM network and performed all communication functions correctly. The module was able to:

- Register on the network
- Dial the programmed phone number
- Establish voice calls

- Maintain active communication
- End calls properly

The SIM800L responded correctly to all AT commands transmitted from the ESP32 microcontroller. The module also demonstrated stable operation when powered using the regulated 4V supply from the buck converter.

4.3.3 LCD Display Performance

The LCD display successfully displayed real-time system status messages during operation. The display improved user interaction by providing useful operational information.

The following messages were displayed during testing:

- “System Ready”
- “Dialing…”
- “Call Connected”
- “Call Ended”

The LCD operated correctly through the I2C communication interface connected to the ESP32.

4.3.4 Audio Communication Performance

The audio communication section consisting of the telephone handset microphone and speaker operated successfully during testing. The handset microphone effectively captured voice input and transmitted audio signals through the SIM800L module. Similarly, the handset speaker produced clear audio output during communication. The use of the intercom handset improved usability and provided communication quality similar to conventional intercom systems. The system achieved real-time two-way communication without the use of an external amplifier circuit.

4.4 Power Supply Performance

The power supply system was implemented using:

- One 18650 rechargeable battery
- TP4056 charging module
- Buck converter

The TP4056 module successfully handled battery charging and protection functions. The buck converter regulated the battery voltage and provided a stable 4V supply required for proper SIM800L operation.

The ESP32 and LCD display operated using the regulated 5V output from the power system.

The implemented power supply prevented:

- Voltage fluctuations
- GSM module restart issues
- Power instability during calls

The rechargeable battery system also improved portability and enabled standalone operation without external power sources.

4.5 System Operation

The GSM-based intercom system operated according to the intended design objectives.

The operational sequence is described below:

- The system powers ON
- ESP32 initializes all connected modules
- LCD displays “System Ready”

- User presses the push button
- ESP32 sends AT commands to SIM800L
- SIM800L dials the programmed phone number
- The receiver answers the call
- Two-way communication begins
- Only the receiver can terminate the call
- System returns to standby mode

The implemented logic ensured that the caller side could not terminate the call accidentally.

4.6 Final Constructed System

The figure below shows the completed GSM-based intercom system developed in this project. The system was assembled on a vero board and integrated inside a portable enclosure together with the telephone handset. The final construction consists of:

- ESP32 microcontroller
- SIM800L GSM module
- LCD display
- Push button
- Rechargeable battery power system
- TP4056 charging module
- Buck converter
- Telephone handset for audio communication

The developed system successfully performs wireless voice communication by placing GSM calls when the push button is pressed. The LCD display provides system status information, while the rechargeable power system ensures portability and uninterrupted operation.



Figure 4.1: Final Constructed GSM-Based Intercom System

4.7 Testing and Results

Several tests were conducted to evaluate the performance of the developed system.

Test Performed	Expected Result	Actual Result
GSM network connection	Successful registration	Successful
Button press detection	Call initiation	Successful
Automatic dialing	Correct number dialed	Successful
LCD display operation	Status display	Successful
Voice communication	Clear audio transmission	Successful
Battery operation	Stable power supply	Successful
Call termination	Receiver ends call	Successful

The testing results confirm that the developed system performed effectively and achieved the intended project objectives.

4.8 Discussion of Results

The developed GSM-based intercom system demonstrated reliable performance throughout testing. The integration of the ESP32 microcontroller with the SIM800L GSM module enabled successful wireless communication over GSM networks. The power supply system proved effective in handling the high current demand of the SIM800L module during call operations. The use of a rechargeable battery system also improved portability and usability. The LCD display enhanced system monitoring and user interaction by displaying operational status messages in real time. The use of a telephone handset improved communication quality and provided a practical intercom-like experience for users.

One major advantage of the developed system is that communication is not limited by physical wiring distance since GSM networks are used for transmission. However, the performance of the system depends largely on GSM signal availability and network quality.

CHAPTER FIVE

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the conclusion of the study, achievements of the project, limitations encountered during implementation, recommendations for future improvements, and the contribution of the project to the field of embedded systems and wireless communication. The chapter summarizes the overall performance of the GSM-based intercom system and evaluates the extent to which the objectives of the study were achieved. The project focused on the design and implementation of a GSM-based intercom system using the ESP32 microcontroller and SIM800L GSM module. The developed system was designed to provide a wireless alternative to conventional wired intercom systems through the use of GSM communication technology. By integrating embedded systems with GSM communication, the developed device was able to support real-time voice communication over long distances without relying on physical cable connections. The successful implementation of the system demonstrates the practical application of wireless communication technologies in solving communication problems associated with traditional wired intercom systems.

5.2 Conclusion

The design and implementation of the GSM-based intercom system was successfully achieved in this project. The developed system successfully integrated the ESP32 microcontroller, SIM800L GSM module, LCD display, rechargeable power supply unit, telephone handset, and push button

interface into a functional wireless communication device capable of supporting real-time voice communication.

The project was carried out to address the limitations associated with traditional wired intercom systems. Conventional intercom systems rely heavily on physical wiring, which increases installation complexity, limits communication range, and reduces flexibility. These systems are also difficult to maintain and expand, especially in large buildings or remote environments where cable installation becomes impractical.

The developed GSM-based intercom system successfully overcame these limitations by utilizing GSM technology for communication. Instead of depending on direct wiring between communication points, the system used GSM cellular networks to establish voice communication over long distances. This significantly improved flexibility and portability.

The ESP32 microcontroller functioned effectively as the central processing and control unit of the system. It successfully processed input signals from the push button, controlled the LCD display, and communicated with the SIM800L GSM module through UART serial communication. The ESP32 maintained stable communication with the GSM module throughout the testing process.
The SIM800L GSM module also performed effectively during operation. The module successfully connected to the GSM network, initiated phone calls, and maintained communication links during testing. The GSM module responded accurately to the AT commands sent by the ESP32 microcontroller and successfully enabled wireless voice communication between users.

One of the major achievements of the project was the successful implementation of real-time two-way voice communication using the telephone handset microphone and speaker. The microphone effectively captured voice signals and transmitted them through the GSM network, while the speaker reproduced incoming audio signals during communication.

The use of a telephone-style handset improved the practicality and usability of the system by providing a communication experience similar to that of a conventional intercom device. In addition, the audio system operated successfully without requiring an external audio amplifier, thereby simplifying the overall system design.

The LCD display unit also contributed significantly to the functionality of the system. The display provided real-time feedback regarding system operation by displaying messages such as “System Ready,” “Dialing,” “Call Active,” and “Call Ended.” This improved user interaction and system monitoring.

The rechargeable power supply system implemented in this project also operated effectively. The power system consisted of an 18650
rechargeable battery, TP4056 charging module, and buck converter for voltage regulation. The SIM800L GSM module requires stable voltage and high current during communication, particularly during call transmission. The implemented power supply system successfully provided stable operation and prevented voltage drops that could result in communication failure or module restart.

Another important achievement of the project was the implementation of system logic that ensured that the call remained active until the receiver terminated it. This made the system behave more like a practical intercom device where communication remains active until intentionally disconnected by the receiver.

The overall testing results confirmed that the developed system operated according to the intended design objectives. The system successfully:

- Detected push button input
- Initiated GSM calls automatically
- Established real-time voice communication
- Displayed system status information
- Maintained stable communication
- Operated using rechargeable battery power

The developed system therefore demonstrates the practical application of GSM communication technology and embedded systems in the development of modern wireless communication devices.

5.3 Achievements of the Study

Several important achievements were accomplished during the course of this project. These achievements demonstrate the successful realization of the project objectives and confirm the effectiveness of the developed system. One of the major achievements of the study was the successful design and implementation of a GSM-based intercom system capable of wireless communication through GSM networks. The developed system successfully demonstrated that

embedded systems and GSM communication technology can be integrated to provide reliable long-distance voice communication.

Another major achievement was the successful interfacing of the ESP32 microcontroller with the SIM800L GSM module. The UART communication established between both devices enabled the ESP32 to transmit AT commands required for call initiation and communication control.

The project also successfully implemented real-time two-way voice communication using a telephone handset microphone and speaker connected directly to the SIM800L module. This enabled the system to function similarly to a conventional intercom device while eliminating the need for wired communication infrastructure.

The implementation of the rechargeable power supply system was also a significant achievement. The power system consisting of the TP4056 charging module, rechargeable battery, and buck converter successfully provided stable voltage and current required for reliable operation of the GSM module and other system components.

The LCD display unit was also successfully integrated into the system to provide operational feedback during system use. The display improved usability by enabling users to monitor system status in real time. Furthermore, all system components were successfully assembled on a vero board and packaged into a compact enclosure together with the handset, resulting in a portable and practical communication device.

5.4 Limitations of the Study

Despite the successful implementation and operation of the system, certain limitations were encountered during the course of the project.

One of the major limitations of the system is its dependence on GSM network availability and signal strength. Since the system operates using GSM communication, poor network coverage may affect call connection quality and communication reliability. In areas with weak network signals, the system may experience delays in call initiation or reduced audio quality.

Another limitation is the relatively high current requirement of the SIM800L GSM module during communication. The module requires stable voltage and current during transmission. If the power supply is not properly regulated, the module may restart or lose network connection during operation.

The system is also limited to communication with a single predefined phone number programmed into the ESP32 microcontroller. Additional software and hardware modifications would be required to support multiple users or dynamic number selection.

Environmental noise may also affect communication quality because the current system does not include advanced noise cancellation or digital audio processing circuitry.

Additionally, the use of GSM communication requires the use of a SIM card and active network subscription, which may result in operational costs depending on network usage.

5.5 Recommendations

Based on the observations made during the implementation and testing of this project, several recommendations are proposed for future improvements and further research.

Future versions of the system can incorporate Wi-Fi communication features to support internet-based communication and remote monitoring capabilities. This would improve flexibility and reduce dependence on GSM networks.

The system can also be improved through the integration of video communication features using camera modules. This would enhance the security and monitoring capabilities of the system.

The use of a custom-designed printed circuit board instead of a vero board is also recommended. A PCB would improve the compactness, durability, and overall appearance of the system while reducing wiring complexity.

Future improvements may also include the addition of a keypad interface to enable users to dial multiple phone numbers instead of relying on a single programmed number.

Advanced audio processing features such as noise filtering, audio amplification, and digital signal processing can also be implemented to improve communication quality.

Furthermore, a mobile application can be developed to allow remote monitoring and control of the intercom system using smartphones or internet-enabled devices.

Battery monitoring circuitry can also be integrated into the system to provide real-time battery status information and improve power management.

The project can also be expanded to support additional automation features such as remote door control, alarm systems, and visitor authentication.

5.6 Contribution to Knowledge

This project contributes to the field of embedded systems and wireless communication by demonstrating the practical implementation of a low-cost GSM-based intercom system using readily available electronic components.

The study also demonstrates how GSM communication technology can be integrated with embedded systems to achieve long-distance wireless voice communication without physical wiring.

The project contributes to knowledge in the areas of:

- Embedded system design
- GSM communication systems
- Wireless communication technology
- Portable electronic system development
- Microcontroller interfacing techniques

The developed system also provides an alternative solution to conventional wired intercom systems and highlights the importance of wireless technologies in modern communication system design.

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